



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY

LONERE - RAIGAD - 402 103

Winter Semester Examination - Dec. - 2019

Branch: B. Pharm. Third Year B. Pharm

Subject: Industrial Pharmacy-I (BP502T)

Date: 18/12/2019

Semester: V

Marks: 75

Time: 3hrs

- Instructions:** i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q. No.1 Multiple Choice Questions (MCQs) = 20 x 1 = 20

(Answer all the questions)

1. Preformulation stability study includes
a) Solid state stability b) Solution state stability c) pH stability d) All of the above
2. Which one of the following solvents is more suitable for determining the partition coefficient that is useful for predicting drug absorption through gastrointestinal tract membrane?
a) Butanol b) Heptane c) Liquid paraffin d) n-octanol
3. The deliquescence is a property in which a solid
a) Absorbs moisture and remains solid b) Absorbs moisture and turns to liquid form
c) Increases the number of water of crystallization d) Loses water of crystallization
4. In the tablet coating process inadequate spreading of the coating solution before drying causes
a) Orange peel effect b) Sticking effect c) Blistering effect d) Picking effect
5. According to BCS classification drugs belonging to Class III shows
a) Low solubility and low Permeability b) Low solubility and high Permeability
c) High solubility and low Permeability d) High solubility and high Permeability
6. Enteric coating is achieved by using
a) Hydroxy Propyl Methyl Cellulose b) Carboxy Methyl Cellulose
c) Cellulose Acetate Phthalate d) Povidone
7. Subcoating is given to the tablet
a) To increase bulk b) To avoid deterioration due to microbial attack
c) To prevent the solubility in acidic medium d) To avoid stickiness
8. The sweetening agent commonly used in chewable tablet is
a) Saccharin b) Marmitol c) Aspartame d) All of the above
9. The measure of cohesive strength of the cross linking that occurs between gelatin molecules and is proportional to the molecular weight of gelatin is called
a) Bloom strength b) Viscosity c) Surface tension d) Iron content
10. In capsules ROTOFIL is used for filling
a) Powders b) Pellets c) Liquids d) Corrosive liquids
11. In the formulation of suspensions for soft gelatin encapsulation base adsorption of the solid to be suspended is expressed as the number of
a) grams of liquid base required to produce a capsulatable mixture when mixed with 1 gm of solid
b) ml of liquid base required to produce a capsulatable mixture when mixed with 1 gm of solid
c) grams of solid base required to produce a capsulatable mixture when mixed

- with 1 gm of solid
- d) mg of liquid base required to produce a capsulatable mixture when mixed with 10 gm of solid
12. The tear secretion contains an antibacterial enzymes known as
 - a) Zymase b) Diastase c) Lysozyme d) Lipase
 13. The growth of large particles at the expense of smaller ones as a result of a difference in the solubility of the particles of varying sizes, is termed as
 - a) Polymorphic transformation b) Crystal habit c) Ostwald ripening d) None of the above
 14. Durability of tablet to combined effects of shock and abrasion is evaluated by using
 - a) Hardness tester b) Disintegration test apparatus c) Friabilator d) Screw gauge
 15. The Wurster process can be used to
 - a) Coat tablets b) Determine the disintegration time c) Tablet compression d) None of the above
 16. The shells of soft gelatin capsule may be made elastic or plastic by addition of
 - a) Sorbitol b) Povidone c) Polyethylene glycol d) Hydroxy Propyl Methyl Cellulose
 17. Green bones are used for the preparation of gelatin of the type
 - a) A b) B c) C d) Both A & B
 18. Which one of the following device is used to increase the efficiency of drug delivery via aerosols?
 - a) Tube spacers b) Metered valve c) Actuator d) Pressure valve
 19. Polymorphism can be detected by all of the following techniques except
 - a) XRPD b) DSC c) IR spectroscopy d) UV spectroscopy
 20. Capillary zone electrophoresis and foaming activity are utilized to determine
 - a) Solubility b) Particle size c) pKa d) None of the above

Q. No.2 Long Answers (Answer 2 out of 3) = $2 \times 10 = 20$

- A] Discuss in detail IPQC tests for tablets.
- B] Differentiate between hard and soft gelatin capsules. Explain the construction and working of rotary die process. Add a note on quality control tests for soft gelatin capsule as per IP.
- C] Explain in detail preformulation study for parenteral drug delivery.

Q. No.3 Short Answers (Answer 7 out of 9) = $7 \times 5 = 35$

- A] Define pellets. Write its application. Discuss extrusion and spheronization.
- B] Solubility and pKa.
- C] Write a note on Glass as a packaging material.
- D] Explain formulation and preparation of cold cream and vanishing cream.
- E] Explain preparation and evaluation of eye ointment.
- F] Write about various properties of gelatin. Explain steps involved in preparation of hard gelatin capsule shell.
- G] Explain Aerosol packaging system.
- H] Enlist quality control test parameters for parenteral preparations. Add a note on pyrogen test and LAL test.
- I] Differentiate between sugar coating and film coating. Explain steps involved in sugar coating.

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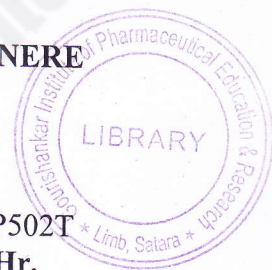
DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 10/02/2023

Course : B. Pharmacy
Subject Name : Industrial Pharmacy-I
Max Marks : 75

Sem: V
Subject Code : BP502T
Duration : 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) What is BCS classification of drug? Enlist each class with example.
- ii) Explain preformulation with definition, goals and objectives.
- iii) Enlist manufacturing defects in tablet.
- iv) Define granulation. Give objectives and enlist methods of granulation.
- v) Define tablet. Enlist types of tablet.
- vi) Write difference between hard gelatin and soft gelatin capsule.
- vii) Write cleaning of containers and closures of parenteral
- viii) Explain any two excipient used in ophthalmic preparation.
- ix) Write a note on pull sealing method.
- x) Give steps involved in pressure and cold filling of aerosol manufacturing

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Enlist and explain pelletization techniques.
- ii) What is parenteral product? Write in detail quality control tests.
- iii) Define pharmaceutical aerosol. Describe the major components of same.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Explain in detail physical properties of new drug molecule.
- ii) Enlist evaluation test of tablet. Explain any four.
- iii) Define excipient. Write excipients used in tablet.
- iv) Explain emulsion and suspension with definition, advantages, limitation and types.
- v) Give the filling methods of hard gelatin capsules.
- vi) Add a note on quality control test of soft gelatin capsule.
- vii) Discuss the preparation and evaluation of eye ointment.
- viii) Describe the formulation and preparation of shampoo.
- ix) Explain quality control test for glass.

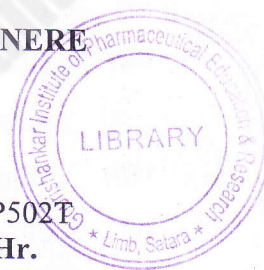
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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination - Winter 2022

Course : B. Pharmacy
Subject Name : Industrial Pharmacy I
Max Marks : 75

Date: 24/07/2023
Sem: V
Subject Code : BP502T
Duration : 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Define Enteric coating and give its advantages
- ii) Explain the significance of isotonicity in parenteral
- iii) Write the BCS classification of drugs
- iv) Write pyrogen test for parenteral
- v) What are pellets? Give advantages and disadvantages of pellets
- vi) Define cosmetics and give classification of cosmetics with examples.
- vii) Write identification test for emulsion.
- viii) What is simple syrup and Justify why preservatives are not added in simple syrup?
- ix) What is tablet tooling ?
- x) What is the difference between flocculated & deflocculated suspension?

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Describe the components , types of aerosol system and Explain the evaluation test for aerosol
- ii) What is tablet? Explain the classification of tablet in detail along with the excipients used in the formulation of tablets.
- iii) Differentiate between Hard gelatin and soft gelatin capsule. Discuss manufacturing and filling methods of hard gelatin capsule

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Define emulsion and explain different methods used in preparation of emulsion
- ii) Write formulation consideration & labeling for ophthalmic preparation
- iii) Elaborate manufacturing consideration of suspension.
- iv) Explain quality control test for packaging material
- v) Write a note on pelletization process.

- vi) Explain the types of tablet coating and defects in coating.
- vii) Discuss Production facilities and Controlled environment required for parenteral preparation
- viii) Write a note on formulation of lipsticks and shampoo
- ix) Elaborate principal areas of preformulation research

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE.

END SEMESTER EXAMINATION WINTER-2023

Course: Bachelor of Pharmacy (B. Pharm)
Subject: INDUSTRIAL PHARMACY-I (THEORY)
Time: Three Hours

Date: 05/01/2024

Semester: Fifth Semester
Subject Code:BP 502 T.
Maximum Marks: 75

- Note: 1. Question No.1 is compulsory
2. Attempt any **two** question from Que. No. 2
3. Solve any **seven** from Que. No. 3

10 x 2 = 20

Que. No. 1. Attempt all questions: -

- Define preformulation. Enlist the physical and chemical properties of drug substance evaluated for preformulation study.
- Define tablet. Mention the reason for its popularity.
- What is the need of coating of tablet and what are the different types of tablet coating?
- Write advantages of pellets
- Draw a schematic labelled diagram of an Aerosol container.
- What are diffusible and indiffusible solids?
- Name the excipients commonly used for making capsule dosage form.
- What is the importance of isotonicity in parenteral formulation with suitable example?
- What are the excipients added in ophthalmic preparations?
- What is important role of packaging materials?

2 x 10 = 20

Que. No. 2. Answer any TWO questions.

- What is aerosol? Explain the different types of aerosols system and quality control parameters for Aerosols.
- Define parenteral formulation along with advantages and limitations. Describe the process of production, facilities and control of parenteral formulation.
- Enlist different excipients and role of excipient used in formulation of tablets. Explain tablet compression and processing problems along with remedies.

7 x 5 = 35

Que. No. 3. Answer any SEVEN questions.

- Describe the various aspects of environmental protection of pharmaceutical products considered in packaging
- Differentiate the cold cream and vanishing cream. Discuss the composition and formulation of vanishing cream.
- Explain the in-process and finished product quality control test for tablet.
- Define coating. Describe steps for sugar coating along with its advantages and disadvantages.
- Describe in brief sterility test for parenterals.
- Explain detailed method of preparation of emulsion with example.
- Describe production process of hard gelatin capsule shell.
- Explain BCS classification system (BCS) along with its application.
- Explain formulation aspects of preparation of ophthalmic formulation with example.



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Pharmacy

Semester: V

Subject Name: Industrial Pharmacy I

Subject Code: BP502T

Max Marks: 75

Date: 15-06-2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions) (10 X 2) = 20

- i) Mention the composition of the glass-I container.
- ii) Write the advantages of pellets over other dosage forms.
- iii) Define Elixers with Examples.
- iv) List out the manufacturing defects of tablets.
- v) Give the identification tests of Emulsions.
- vi) Define bloom strength.
- vii) What is BCS System?
- viii) Differentiate between Hard gelatin Capsules and Soft gelatin capsules.
- ix) Enlist Evaluation Test for Aerosols.
- x) What are hydrates and solvates?

Q.2. Long Answers (Answer 2 out of 3) (10 X 2) = 20

- i) Classify granulation techniques. Discuss the wet granulation method along with the equipment used in the each step.
- ii) Write in details Quality control tests of Parenterals.
- iii) Explain about the Formulation of hard gelatin capsule

Q.3. Short Answers (Answer 7 out of 9) (5 X 7) = 35

- i) Discuss the steps involved in sugar coating with ingredients used in each step.
- ii) Explain the steps involved in manufacturing of hard gelatin capsules shells.
- iii) What are excipients used in liquid formulation
- iv) Write a note on Formulation of lyophilized sterile products.
- v) How lipsticks are evaluated.
- vi) Write a note on Hot Melt Extrusion
- vii) Explain in short about Pressure filling and cold filling of Aerosols.
- viii) Write a note on Ophthalmic Preparations.
- ix) Elaborate the parenteral additives with their examples.

*** END OF THE PAPER ***

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